Angle relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies angle relationships to solve problems, including those related to transversals on sets of parallel lines **MA4-ANG-C-01**

Content

Apply the language, notation and conventions of geometry

- Use appropriate terminology and conventions to define, label and name points, rays, lines and intervals using capital letters

Example(s):

Describing the similarities and differences between the following examples:

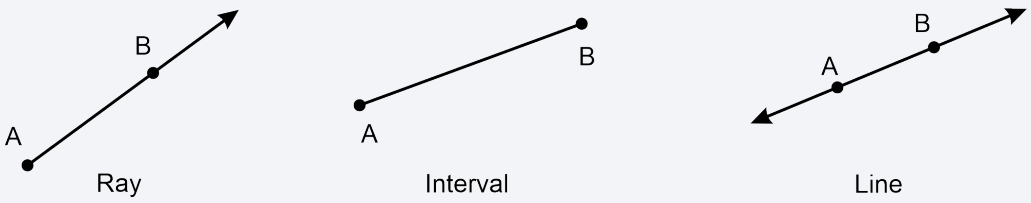
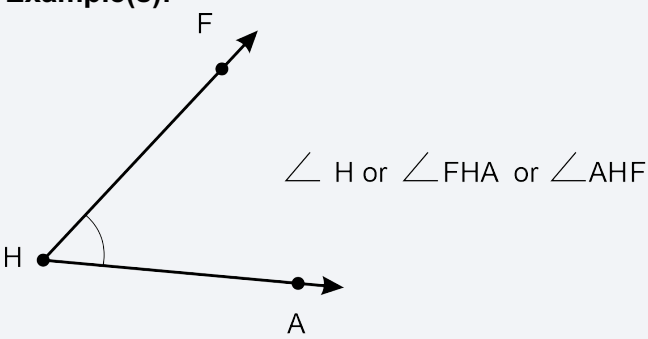


Image long description: The first diagram is a ray starting at point A and continuing through point B. The second diagram shows an interval of a line starting at point A and ending at point B. The third shows a line that goes through points A and B and continues through both.

- Identify and label the vertex and arms of an angle with capital letters
- Use appropriate conventions to label and name angles

Example(s):



- Use common conventions to indicate right angles, equal angles and intervals on diagrams

Example(s):

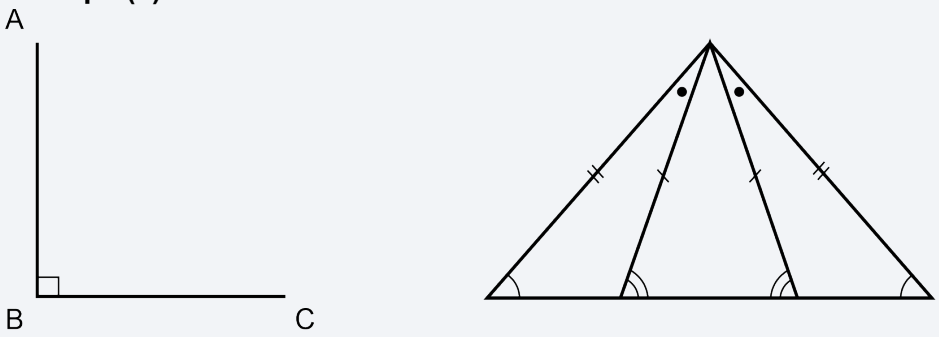


Image long description: The first diagram is a right angle through the points ABC. The second diagram has one isosceles triangle inside another isosceles triangle showing equal sides and equal base angles.

Identify geometrical properties of angles at a point

- Identify right angles, straight angles, angles of complete revolution and vertically opposite angles
  UuM8
- Apply the terms *complementary* and *supplementary* to a pair of angles adding to *90°*and *180°*, respectively

Example(s):

Identifying complementary and supplementary angles arising from right angles and straight angles. $\angle ABC$ and $\angle CBD$ are complementary since $\angle ABD$ is a right angle. $\angle FHG$ and $\angle JHG$ are supplementary since $\angle FHJ$ is a straight angle.

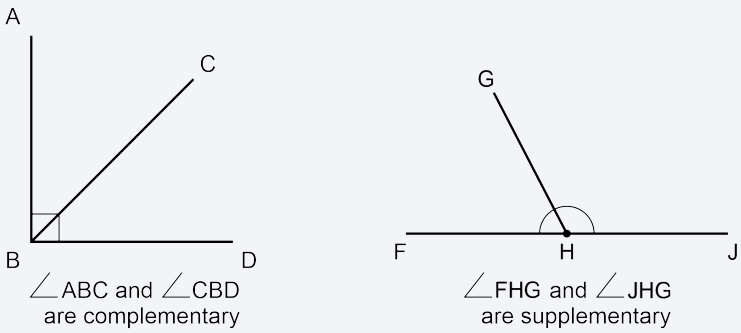


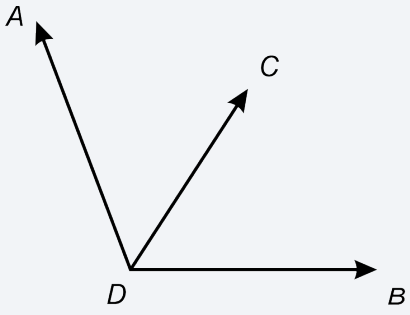
Image long description: The first diagram shows right angle *ABD* dissected by line *BC* forming complementary angles. The second diagram shows straight angle *FHJ* with line *GH* in the middle at a diagonal forming supplementary angles.

UGP5

- Apply the term *adjacent angles* to a pair of angles with a common arm and common vertex

Example(s):

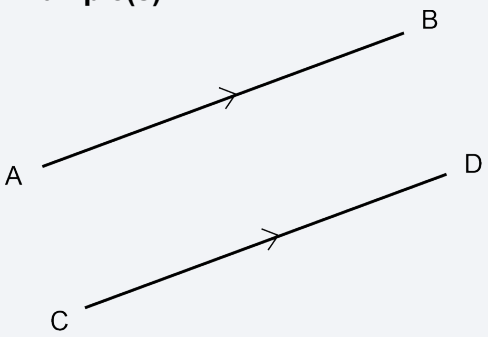
$\angle ADC$ is adjacent to $\angle BDC$ and reflex $\angle ADB$.



Identify and describe corresponding, alternate and co-interior angles when 2 straight lines are crossed by a transversal, including parallel lines

- Identify and describe perpendicular lines using the symbol for *is perpendicular to* (⊥)
  UuM8, **UGP6**
- Apply the common conventions to indicate parallel lines on diagrams

Example(s):

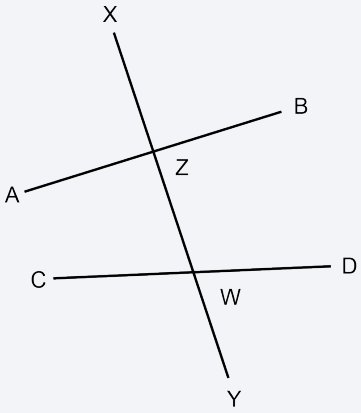


- Identify and describe pairs of parallel lines using the symbol for *is parallel to* (∥)
  UGP6

- Identify and define transversals, including transversals of parallel lines

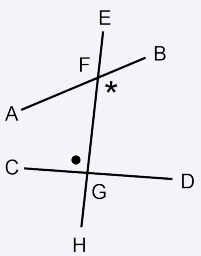
Example(s):

Transversal XY cuts the lines AB and CD at Z and W.

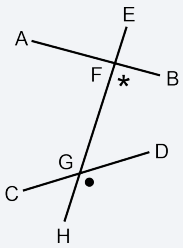


- Identify, name and measure alternate angle pairs, corresponding angle pairs and co-interior angle pairs for 2 lines cut by a transversal

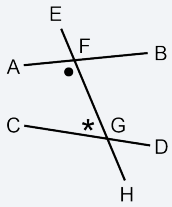
Example(s):



∠CGF and ∠BFG are alternate angles



∠HGD and ∠BFG are corresponding angles

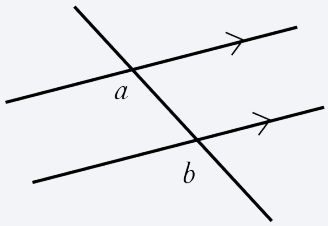


∠AFG and ∠CGF are co-interior angles

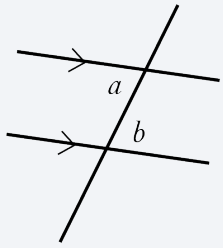
UGP6

- Verify and identify corresponding angles and alternate angles as equal, and co-interior angles as supplementary, when a pair of parallel lines is cut by a transversal

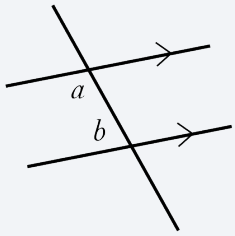
Example(s):



$a = b$



$a = b$



$a + b = 180^\circ$

Image long description: A corresponding angle and alternate interior angles where angle *a* and angle *b* are equal. A co-interior angle where angle *a* plus angle *b* equals 180 degrees.

UGP6

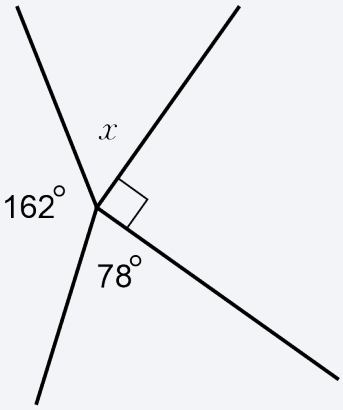
- Justify that 2 lines are parallel by using properties of alternate, corresponding or co-interior angles on parallel lines
  UGP6

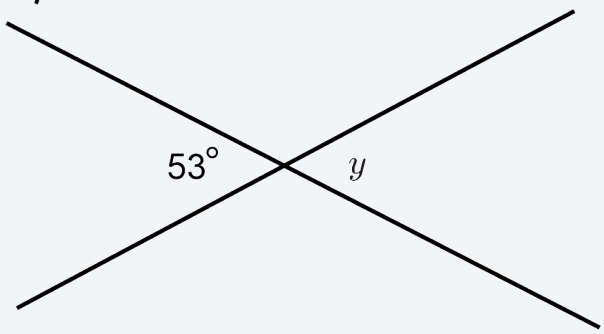
Solve numerical problems involving angles using reasoning

- Apply the knowledge of angle relationships including angles at a point to find the sizes of unknown angles embedded in diagrams and give reasons

Example(s):

Finding the size of the unknown angle and give reasons:





$x + 162° + 90° + 78° = 360°$ (angles at a point)

$\therefore x = 30°$

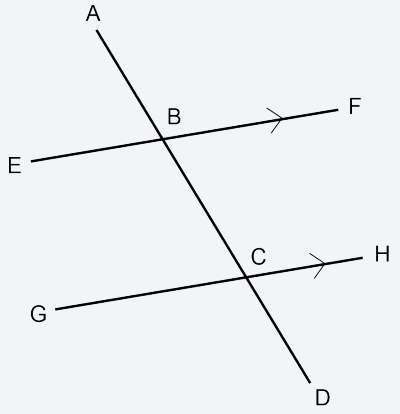
$y = 53°$ (vertically opposite angles)

UGP5, NPA6

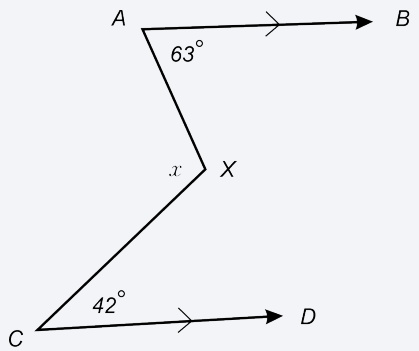
- Apply the knowledge of angles associated with parallel lines to find the sizes of unknown angles embedded in related diagrams and give reasons

Example(s):

If $\angle BCH = 114°$, find the size of $\angle ABF$ and $\angle EBC$ and give reasons.



Finding the value of x in the diagram below and give reasons:



Construct a line XY through X parallel to AB and CD.
$63^\circ + \angle AXY = 180^\circ$ (co-interior angles, $AB \parallel XY$)
$\therefore \angle AXY = 117^\circ$
Similarly, $\angle YXC + 42^\circ = 180^\circ$
$\therefore \angle YXC = 138^\circ$
$x + 117^\circ + 138^\circ = 360^\circ$ (angles at a point)
$\therefore x = 105^\circ$

UGP6, NPA6

Properties of geometrical figures

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and applies the properties of triangles and quadrilaterals to solve problems **MA4-GEO-C-01**

Related Life Skills outcomes: MALS-GEO-01

Content

Classify triangles according to their side and angle properties

- Label triangles using appropriate text and symbols

Example(s):

Triangle ABC can be written as $\triangle ABC$.

- Classify and describe types of triangles based on their properties, including acute-angled, right-angled, obtuse-angled, equilateral, isosceles and scalene triangles

Example(s):

Recognising that a given triangle may belong to more than one class such as right-angled scalene triangles and obtuse-angled isosceles triangles.

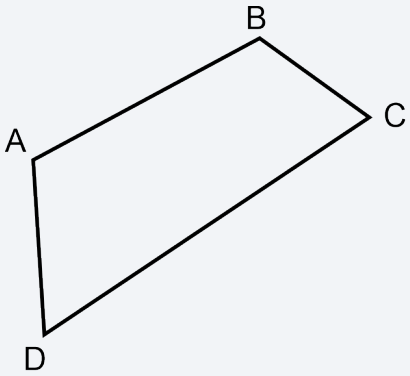
UGP6

Classify quadrilaterals and describe their properties

- Identify quadrilaterals using naming conventions

Example(s):

Recognising that quadrilateral ABCD is the same as ADCB but not the same as ACBD.

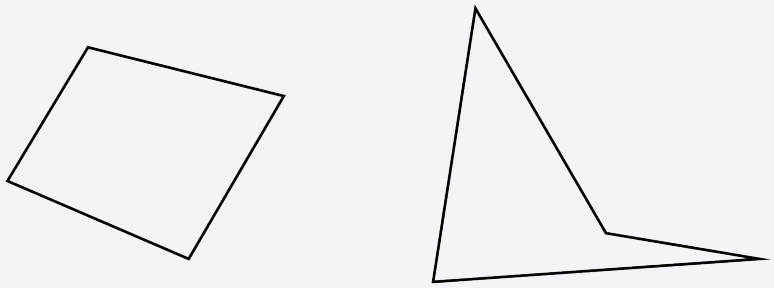


UGP4

- Distinguish between convex and non-convex quadrilaterals

Example(s):

Non-convex (also called concave) quadrilaterals have one internal reflex angle.



convex concave

UGP4

- Verify and describe the properties of the special quadrilaterals which include parallelograms, rectangles, rhombuses, squares, trapeziums and kites

Example(s):

Properties of special quadrilaterals:
Opposite sides are parallel.
Opposite sides are equal.
Adjacent sides are perpendicular.
Opposite angles are equal.
Diagonals are equal.
Diagonals bisect each other.
Diagonals bisect each other at right angles.
Diagonals bisect the angles of the quadrilateral.

UGP4

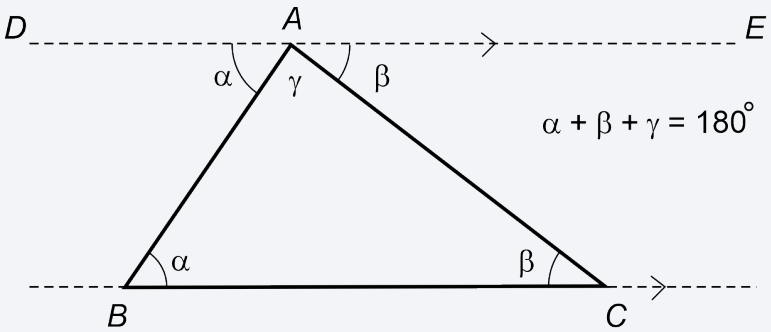
- Identify and label the properties of the special quadrilaterals using appropriate conventions
  UGP4
- Classify quadrilaterals based on their properties
  UGP4
- Justify why some quadrilaterals may be classified as more than one type of quadrilateral
  UGP4

Apply the properties of triangles and quadrilaterals

- Prove that the interior angle sum of a triangle is 180° with or without digital tools

Example(s):

Proving the interior angle sum of a triangle is 180°:



Construct DE parallel to BC.

$\angle DAB = \angle ABC = \alpha$ (alternate angles, $DE \parallel BC$)

$\angle EAC = \angle BCA = \beta$ (alternate angles, $DE \parallel BC$)

$\alpha + \beta + \gamma = 180^\circ$ (straight angle)

$\therefore \angle ABC + \angle BCA + \angle CAB = 180^\circ$

Image long description: Triangle *ABC* with interior angles alpha, beta and gamma. Line *DE* is constructed parallel to *BC* at point *A* forming a straight angle of alpha, gamma and beta with a sum of 180 degrees.

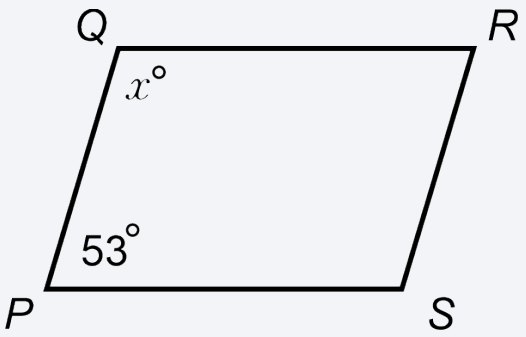
UGP6

- Prove that any exterior angle of a triangle equals the sum of the 2 interior opposite angles
  UGP6
- Apply the angle sum of a triangle to prove that the angle sum of a quadrilateral is 360°
  UGP6

- Apply the properties of triangles and quadrilaterals to determine unknown sides and angles to solve problems, giving reasons

Example(s):

Finding the value of x given $PQRS$ is a parallelogram:



$QR \parallel PS$ (opposite sides of a parallelogram are parallel)

$x + 53^\circ = 180^\circ$ (co-interior angles are supplementary, $QR \parallel PS$)

$\therefore x = 127^\circ$

UGP6, NPA6